

Direct device access from the SmartNIC towards datacenter disaggregation

Master's thesis meeting : week 9

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SOTA updates

- Found additional papers for storage offloading / handling

My Library Master's thesis To read and eval... seminar My Publications Duplicate Items Unfiled Items Trash	Title	Creator	
	>  An RDMA-First Object Storage System with SmartNIC Offload	Zhu et al.	
	>  Characterizing Off-path SmartNIC for Accelerating Distributed Systems	Wei et al.	
	>  LeapIO: Efficient and Portable Virtual NVMe Storage on ARM SoCs	Li et al.	
	>  Poby: SmartNIC-accelerated Image Provisioning for Coldstart in Clouds	Chang and Zhu	
	>  SmartDS: Middle-Tier-centric SmartNIC Enabling Application-aware Message Split for Disaggregated Block Storage	Zhang et al.	

SPDK progress

- Finished setup on Frodo
- Tried examples scripts like *bdevperf*

Objectives

- Read for inspiration on papers
- Split data like TCP/IO but for NVMe (still a fuzzy idea)
 - Seems like what could be gain in flexibility would be lost in performance and vice versa
- **I'm lost**

Poster

- Context : speak about resource stranding
- Problem : not found yet
- Approach :
 - data transfer with SPDK on local server
 - transfer on the network with DPDK
 - If enough time, encryption

Conclusion

- Working on paper creation
- Added a few papers to know more about storage offloading

TODOs for week 11

- More SPDK scripts
- I hope to find what the thesis problem is by week 11
- Investigate NVMe Computational SSD (it is interesting for our storage disaggregation ?)

That's all for today !